Jialin Li

jialin.li@nyu.edu | Homepage | Google Scholar

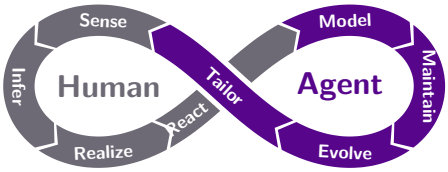
Education

New York University — Sept. 2024 – May 2029 (Expected)
Ph.D. in Computer Science, NYU Abu Dhabi Global Ph.D. Fellow — *New York, NY, USA*
Advisor: Hanan Salam

New York University Shanghai — Sept. 2019 – Jan. 2023
B.S. in Data Science — *Shanghai, China*

Research Interests

- **Research Focus:** People-Aware LLM Agents, building and benchmarking personal agents that model individual affect and personality, maintain context-dependent interaction preferences over time, and evolve their models and harnesses from user feedback.
- **Long-Term Goal:** Co-Evolving Human-AI Partnerships where every interaction turns into learning signals that tailor the agent's harness to the user while preserving full human agency.



Sense / Infer → Multimodal Cues & Latent Preferences
Realize / React → Adapted Replies / Tool Calls & User Reactions
Model / Maintain → Preference Evidence & Evolving Memory States
Evolve / Tailor → Post-Training Alignment & Personalized Harness

① Human Signals
Q: How to model what makes each person *distinct*?
How: Personality-aware & per-user multimodal affect modeling
MOCSE · PersonAug · PAC Survey

→

② Preference States
Q: How can agents *understand, maintain & realize* how you prefer to interact?
How: Context-resolved preference adaptation & long-horizon memory
PrefIx · MemPABench

→

③ Adaptive Harnesses
Q: How can models & harnesses *co-evolve* from user feedback?
How: Turning interaction reactions into alignment & harness updates
Alignment & Harness Evolution

Experience

Tencent Hunyuan, Foundation Model Team — May 2026 – Sept. 2026
Algorithm Research Intern, In-Game Companion Agents — *China*

- **Interactive Data Synthesis & Evaluation ③:** To overcome scarce state-grounded dialogue, context drift, and unnatural TTS cadence, developed a **two-sided in-game synthesis pipeline** integrating **player personas** and NPC **cognitive profiles** with state-alignment rules, synthesizing **10K+** role-consistent multi-turn dialogues. Implemented a two-layer rubric evaluator applying **gating metrics** (safety, logic) as hard filters to discard invalid samples and **uplift metrics** (spoken style, plot progression) for fine-grained ranking. Calibrated the evaluator against human ratings to standardize data filtering and RL reward modeling.
- **Real-Time State & Layered Memory ②:** Rapid PUBG state changes and **asynchronous** reasoning often led to stale context and module-level hallucinations. Consolidated game state, match events, and dialogue into a shared, **incrementally updated structured context**, cutting mismatched triggers by **8%**. Designed a lifecycle-based **L0–L4 layered memory** framework with automatic summarization, constructing targeted training data to optimize memory recall and natural follow-up generation.
- **Implicit-Feedback Reward & Multi-Turn RL ③:** Single-turn RL failed to optimize long-horizon conversational goals over multi-turn dialogues. Modeled implicit player preferences across **behavioral signals** (mic usage, chat continuity) and **subjective states** (sentiment, interaction tone) to calibrate a **multi-objective reward model**. Executed staged training: SFT for prompt compression and persona style, followed by **offline PPO and online GRPO** for boosting composite reward scores.
- **Attribution & Data Flywheel ③:** Built an online bad-case analysis pipeline processing **20K** live sessions a day. Replaced single-point scoring with multi-stage analysis to isolate root causes by **system layer**, cutting regression evaluation cycles by **50%+**. Closed a **feedback → attribution → retraining data flywheel** for targeted data iteration, supporting a product serving **8M total users** with a peak DAU of **500K**.

Publications

ACL Findings 2026 **PrefIx: Understand and Adapt to User Preference in Human-Agent Interaction**
J. Li, Z. Chen, H. Luo, H. Salam. Code ②

- **Motivation:** LLM agents can complete tasks correctly yet frustrate users through excessive confirmations or misaligned pacing; existing benchmarks measure task accuracy, not how agents interact.
- **Approach:** Introduced ***interaction preferences*** and **Interaction-as-a-Tool (IaaT)**, which models interaction behaviors (confirm, disambiguate, abort, escalate) as structured tool calls so they fit existing tool-use evaluation. Built 31 preference settings over 14 attributes in four dimensions (transparency, interaction pace, strategic proactiveness, robustness) with LLM-simulated users, scored on seven UX dimensions by a four-LLM judge panel.
- **Findings:** Preference-aware agents keep task accuracy while improving all seven UX dimensions (+7.6% average UX, +18.5% preference alignment); transparency preferences are the easiest to align, robustness the hardest.

Under Review **Why Bother Me Again? Benchmarking Personal Agent Memory for Interaction Preferences**
J. Li, Z. Chen, H. Luo, H. Salam. Code Webpage ②

- **Motivation:** Building on PrefIx, an interaction preference only helps long-term if the agent recalls it in context and updates it as users change, yet existing memory benchmarks miss this by not testing interaction preferences.
- **Approach:** Built MemPABench, an interactive environment for **on-policy memory evaluation**: four grounded simulated users hold 14 interaction preferences across work and personal contexts and react to the agent over 99 sessions spanning four months, with five event-driven preference changes per user. Scored preference selection, delivered behavior and user-perceived quality for four memory systems on four LLM backbones.
- **Findings:** Retrieval-based memory (Mem0, MemOS, RAG) only matches no memory; file memory helps (59.2%) but misapplies preferences across contexts; even with the correct preference, agents realize it only 59% of the time.

ACII 2025 (Oral) **Improving Personalisation in Valence and Arousal Prediction using Data Augmentation**
J. Li, M. Nwadike, H. Salam. ①

- **Motivation:** Per-person valence and arousal models have little data for each individual.
- **Approach:** Proposed a data-driven, **sample-level augmentation** that, for each target segment, retrieves the most similar segments from a pooled multi-person dataset, improving CCC on a personalization benchmark without changing model architectures.

ACM MM 2025 Workshop **Personality-Aware Engagement Prediction in Online Learning**
J. Li, G. Sharma, H. Salam. ①

- **Motivation:** Vision-based engagement prediction ignores individual differences, and existing datasets are mostly lab-collected, one-dimensional, and lack self-reports and personality.
- **Approach:** Built MOCSE, an in-the-wild dataset of Zoom lectures (72 students, 36.5 hours) with self-reported and observer engagement labels plus Big Five personality; conditioning VLMs on personality improves prediction.

arXiv 2023 **Beyond One-Size-Fits-All: A Survey of Personalized Affective Computing in Human-Agent Interaction**
J. Li, M. Elgarf, A. Waleed, H. Salam. ①

- First survey of personalized affective computing, with a taxonomy of data- and model-level approaches.

Skills

Evaluation & Data: Interactive benchmark and environment design, LLM-as-a-judge, multi-turn data synthesis
Post-Training: SFT, reward modeling, PPO, GRPO
Programming: Python, Rust, Shell, SQL; PyTorch, Transformers; Docker, Git, Linux, WandB; Claude Code, Codex

Honors & Awards

NYU Abu Dhabi Global Ph.D. Fellowship, New York University 2024 –
University Honors Scholar, New York University 2023
Dean's List of Academic Year, New York University 2021 – 2022
Dean's Undergraduate Research Fund, New York University 2020 – 2021

Service

Reviewer: ACM MM 2025, SuSS 2025, ACL 2026, ICLR 2026, IEEE TAC, EMNLP 2026
Outreach: Association for the Advancement of Affective Computing (AAAC) DEI Committee, 2025